UNIVERSITY OF BOTSWANA

DEPARTMENT OF MATHEMATICS

2002/03 — END SEMESTER 1 EXAMINATIONS

COURSE NUMBER **MAT111**

DURATION: 2 HRS

DATE: **DECEMBER 2002**

TITLE OF PAPER

INTRDUCTORY CONCEPTS OF MATHEMATICS

SUBJECT: **MATHEMATICS**

TITLE OF EXAMINATION: **BSc. I**

INSTRUCTIONS

ANSWER ALL TEN (10) QUESTIONS FROM SECTION A AND ANY TWO 2 QUESTIONS FROM SECTION B

NO. OF PAGES

2

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SECTION A

Simplify

$$\{\frac{x^2-y^2}{x}\}\{\frac{y^2}{x^3-y^3}\}\{\frac{x^2+xy+y^2}{x^2+2xy+y^2}\}$$

2. Solve the inequality $|3x+4|<5$.

Solve the equation

$$2(10^{3x})-10^{2x}$$

4. '

$$x^2+3x+1$$

are α and β. Find the equation whose roots are $2\alpha, 2\beta$,

5. When the polynomial $f(x)=x^2+ax+b$ is divided by $x-1,\ x+1$ it leaves the remainders 1,25 respectively. Find the constants a and b.

$= \sqrt{x^2-4}$

7. Assuming the expansions of $\sin 2\theta$ and $\cos 2\theta$, prove that $\tan 2\theta =$

8. Solve the quadratic equation $x^2+x-6=0$ by completing the square and hence find the minimum point of the graph of the function $y=x^2+x-6$.

9. Given $\sin\theta=\frac{3}{5}$, find $\cos 2\theta$

$$x^2+y^2-4x+6y-12=0$$

SECTION B

(a) Given the function $f(x) = \dfrac{x+3}{x-1}$, find the inverse function $f^{-1}(x)$. Determine the intercepts and asymptotes the inverse function.

(b) Find the maximum value of the function $f(x) = 2x - x^2 + 8$. Hence sketch the graph of the function.

12. Show that the equation

$$x \quad - 4x + 4y^2 - 8y + 4 \quad 0$$

represents the equation of an ellipse. Determine the vertices, directrix lines and foci .

13. Let $f(x) = x^4 + 4x^3 - x^2 - 16x - 12$

(a) Sketch $y = f(x)$, indicating the x and y intercepts.

(b) Hence sketch $y = f(x+2)$ indicating the x intercepts.

14 (a) Find $\sin \frac{\pi}{12}$, using $\sin(x - y) = \sin x \cos y - \sin y \cos x$.

(b) Find $\sin \frac{\pi}{12}$, using the half angle formula.

(c) Prove that the answer to part a) and part b) are the same number